iLearn Practice Exam

10BR · Chapter / Topic 3

Cambridge IGCSE Computer Science 0478

Paper 1 Computer Systems

Chapter 3: Hardware

Name: _____

Class: _____

Date: _____

Time allowed: 1 hour 30 minutes

TOTAL MARKS: 75

Marks obtained: ..... / 75

INSTRUCTIONS

· Answer all questions.
· Write your answers in the spaces provided on this paper, using a black or dark blue pen.
· Do not use an erasable pen or correction fluid.
· No calculators are allowed.
· Do not write anything in the margins.

INFORMATION

· The total mark for this paper is 75.
· The number of marks for each question or part question is shown in brackets [ ].
· This paper contains 9 questions, all drawn from Cambridge past papers on this chapter.
· After you finish, your teacher will mark your work against the official mark scheme.

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1 Secondary storage devices are used to store data in a computer.

(a) Circle **three** components that are secondary storage devices.

central processing unit (CPU) compact disk (CD)

hard disk drive (HDD) random access memory (RAM) read only memory (ROM)

register sensor solid-state drive (SSD)

[3]

(b) Tick (✓) **one** box to show which statement about secondary storage is correct.

A It is directly accessed by the CPU. ☐

B It is magnetic storage only. ☐

C It is used to permanently store software and data files. ☐

D It is volatile. ☐

[1]

2 A device can be given an internet protocol (IP) address. This can be an IPv4 or IPv6.

(a) Give **one** similarity between IPv4 and IPv6.

.....

..... [1]

(b) Describe **two** differences between IPv4 and IPv6.

1 .....

.....

.....

.....

2 .....

.....

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.....

[4]

(c) A web page is requested using an IP address.

(i) Identify the system that stores a database of uniform resource locators (URLs) and their corresponding IP addresses.

..... [1]

(ii) Identify the software that sends a request to the IP address to obtain the web page data.

..... [1]

3 A user's computer has a central processing unit (CPU) that has a clock speed of 2 GHz.

She wants to change it to a CPU that has a clock speed of 3 GHz.

(a) **(i)** State what is meant by clock speed.

.....

..... [1]

(ii) Explain the effect this change will have on the performance of the CPU.

.....

.....

.....

..... [2]

(b) The CPU contains a memory address register (MAR).

Describe the role of the MAR in the fetch–decode–execute cycle.

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.....

.....

..... [2]

(c) The CPU has a list of all the machine code commands it can process.

State the name of this list of commands.

..... [1]

4 A company uses both solid-state and optical secondary storage.

(a) Explain why a computer needs secondary storage.

.....

.....

.....

..... [2]

(b) Describe **three** differences between solid-state and optical storage.

1 .....

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.....

.....

2 .....

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3 .....

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[6]

5 A central processing unit (CPU) performs the fetch–decode–execute (FDE) cycle.

(a) Give the name of **two** registers that are used in the fetch stage of the cycle.

1 .....

2 .....
[2]

(b) Describe what happens at the decode stage of the cycle.

.....

.....

.....

.....

.....

..... [3]

(c) Give **one** register in the CPU that is used in the execute stage of the cycle.

..... [1]

(d) Buses are used in the CPU to transmit data through the FDE cycle.

Circle **three** buses that are used in the CPU.

fetch	address	register	execute
data	decode	calculation	
central	value	binary	control

[3]

(e) A user changes their CPU from one with a dual core and a clock speed of 2.4 GHz to one with a dual core and a clock speed of 3.5 GHz.

Explain the effect this change will have on the performance of the CPU.

.....

.....

.....

..... [2]

6 An instruction is fetched from random access memory (RAM) into the memory data register (MDR) to be decoded.

(a) Identify **two** other registers that are used in the fetch stage of the cycle.

1 .....

2 .....

[2]

(b) Complete and annotate the diagram to show how the data is decoded once it has been fetched into the MDR.



[4]

7 A student's computer has primary storage. One example of primary storage is random access memory (RAM).

(a) Identify **one** other example of primary storage.

..... [1]

(b) Explain why RAM is an example of primary storage and **not** secondary storage.

.....

.....

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.....

..... [3]

(c) The student is editing a large video file for a project and during this process the RAM becomes full. This makes the computer crash.

Describe how virtual memory could be used in this process to stop the computer crashing.

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..... [3]

(d) The computer uses solid-state (flash memory) storage.

(i) Give **two** features of solid-state (flash memory) storage.

1 .....

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2 .....

..... [2]

(ii) Give **two** examples of solid-state (flash memory) storage.

1 .....

2 ..... [2]

8 A computer with a Von Neumann architecture has a central processing unit (CPU) that performs the fetch–decode–execute (FDE) cycle.

(a) Complete and annotate the diagram to show how instructions are decoded and executed.



[5]

(b) The CPU is dual core and has a processor speed of 2.4 GHz.

Describe what is meant by a core.

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.....

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..... [2]

(c) Explain **two** ways that the performance of the CPU could be increased.

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[4]

9 A smartphone has a quad-core 2.4 GHz processor.

(a) The processor performs the fetch-decode-execute (FDE) cycle.

(i) The first register used in the FDE cycle stores the location in random access memory (RAM) of the next instruction.

Identify the name of this register.

..... [1]

(ii) Describe the actions that take place in the decode stage of the FDE cycle.

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..... [3]

(iii) Give the name of **one** component that is used in the execute stage of the FDE cycle.

..... [1]

(b) Explain the advantages of the smartphone having a quad-core processor instead of a dual-core processor.

.....

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..... [2]

(c) The smartphone has built-in input and output devices.

(i) Identify **two** input devices that could be built into the smartphone.

1 .....

2 .....
[2]

(ii) Identify **two** output devices that could be built into the smartphone.

1 .....

2 .....
[2]

END OF QUESTION PAPER